

CHEMISTRY

TARGET : JEE- Advanced

CAPS-8**COORDINATION COMPOUND****SINGLE CHOICE QUESTIONS**

- Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
 - Co(III) is stabilized in presence of weak field ligands, while Co(II) is stabilized in presence of strong field ligand.
 - Four coordinated complexes of Pd(II) and Pt(II) are diamagnetic and square planar.
 - $[\text{Ni}(\text{CN})_4]^{4-}$ ion and $[\text{Ni}(\text{CO})_4]$ are diamagnetic tetrahedral and square planar respectively.
 - Ni^{2+} ion does not form inner orbital octahedral complexes.

(A) TFTF (B) TTTF (C) TTFT (D) FTFT
- Which of the following order of CFSE is incorrect?

(A) $[\text{Co}(\text{en})_3]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$

(B) $[\text{PtCl}_4]^{2-} > [\text{PdCl}_4]^{2-} > [\text{NiCl}_4]^{2-}$

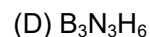
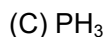
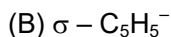
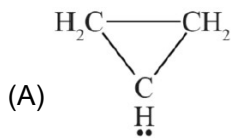
(C) $[\text{Ni}(\text{DMG})_2] < [\text{Ni}(\text{en})_2]^{2+}$

(D) $[\text{Co}(\text{ox})_3]^{3-} < [\text{Co}(\text{en})_3]^{3+}$
- Select the correct code of TRUE and FALSE for given statements:
 - Peroxide ion as well as dioxygen molecule both are paramagnetic species.
 - In set of isomers, $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$, both compounds can easily lose water molecule on treatment with conc. H_2SO_4 .
 - During transformation $\text{NO} \rightarrow \text{NO}^+$, bond length and magnetic behavior decreases.
 - An ether is more volatile than alcohol both having same molecular formula.

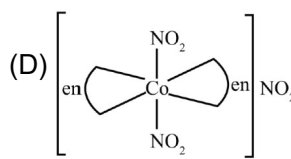
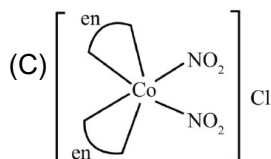
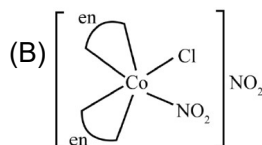
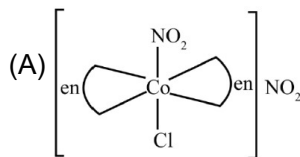
(A) FFTT (B) FTFT (C) FTTT (D) TFFT
- The total possible coordination isomers for the following compounds respectively are:

$[\text{Co}(\text{en})_3][\text{Cr}(\text{C}_2\text{O}_4)_3]$ $[\text{Cu}(\text{NH}_3)_4][\text{CuCl}_4]$ $[\text{Ni}(\text{en})_3][\text{Co}(\text{NO}_2)_6]$

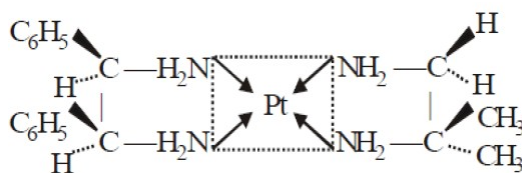
(A) 4, 4, 4 (B) 2, 2, 2 (C) 2, 2, 4 (D) 4, 2, 4
- Ligands are broadly classified into two classes classical and non-classical ligands, depending on their donor and acceptor ability. Classical ligands form classical complexes while non-classical ligands form non-classical complexes. Bonding mechanism in non-classical is called synergic bonding.

Which is not π -acceptor ligand?

6. In complexes of weak field ligands, $\Delta_0 < P$ (Pairing energy), the energy difference between t_{2g} and e_g sets is relatively less. Under the influence of strong field ligands, $\Delta_0 > P$ (Pairing energy), the energy difference between t_{2g} and e_g sets is relatively high. Select the correct statement regarding $[\text{Cr}(\text{en})_2\text{Cl}_2]^+$ and $[\text{Co}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^-$ complex ions:
- (A) Both are equally stable complexes
 (B) Both have equal number of stereoisomers
 (C) Both are diamagnetic complexes
 (D) (a) and (c) both
7. Which of the following is correct?
- (A) When FeCl_3 is treated with $\text{K}_3[\text{Fe}(\text{CN})_6]$, a brown colored complex is formed
 (B) When Fe^{3+} salt treated with CH_3COONa solution, yellow ppt is formed, i.e., $[\text{Fe}_3(\text{OH})_2(\text{CH}_3\text{COO})_6]^+$
 (C) Fe^{3+} salt forms $\text{Fe}_2[\text{Fe}(\text{CN})_6]$ on treating with $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (D) Fe^{3+} salt forms greenish coloured precipitate with $(\text{NH}_4)\text{SCN}$
8. A coordination complex with empirical formula $\text{Co}(\text{en})_2(\text{NO}_2)_2\text{Cl}$ is available in a number of isomeric forms. One of them (A) reacts with both AgNO_3 and ethylenediamine, while another form (B) reacts with only AgNO_3 but not with ethylene diammine. At the same time, another isomer (C) reacts with ethylene diammine but not with AgNO_3 while another isomer (D) neither reacts with AgNO_3 nor with ethylene diammine. Now answer the following questions on the basis of information given above Isomer A is



9. Select the correct statement about given square planar complex.



- (A) It has no geometrical isomer
 (B) It is optically active because it does not have plane of symmetry
 (C) It is optically inactive because square planar complex have plane of symmetry
 (D) It is optically active because it has symmetric carbon.

10. Total number of stereo isomers of $[\text{Co}(\text{acac})_2\text{BrCl}]^-$ are:
 (A) 4 (B) 3 (C) 6 (D) 2
11. Which of the following is correct IUPAC name of any complex compound?
 (A) Tris(acetylacetonato)iron(III) chloride
 (B) Hexachloroplatinum(IV) tetraamminedicyanoplatinate(IV)
 (C) Amminebromochloromethylamineplatinum(II)
 (D) Cis-dichloro(ethylenediamine)platinum(IV)
12. 0.001 mol of $\text{Co}(\text{NH}_3)_5(\text{NO}_3)(\text{SO}_4)$ was passed through a cation exchanger and the acid coming out of its required 20 ml of 0.1 M NaOH for neutralisation. Hence, the complex is –
 (A) $[\text{CoSO}_4(\text{NH}_3)_5]\text{NO}_3$ (B) $[\text{Co}(\text{NO}_3)(\text{NH}_3)_5]\text{SO}_4$
 (C) $[\text{Co}(\text{NH}_3)_5](\text{SO}_4)(\text{NO}_3)$ (D) None of these
13. **Statement-I:** C–C bond length in zeise's salt is same as ethylene.
 Because
Statement-II: Double bond is shorter as compared to single bond.
 (A) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.
14. Consider the following statements
S₁: $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ and $[\text{CoF}_6]^{3-}$ are paramagnetic having four, five and four unpaired electrons respectively.
S₂: Low value of formation constant of a complex indicates its high thermodynamic stability.
S₃: The crystal field splitting Δ_o , depends upon the field produced by the ligand and charge on the metal ion.
 Arrange in the order of true/ false.
 (A) T T T (B) T F T (C) F T F (D) T F F

MULTIPLE CHOICE QUESTIONS

15. Consider the following two carbonyl compounds
 (i) $[\text{Tc}(\text{CO})_6]^+$ and (ii) $[\text{Nb}(\text{CO})_6]^-$
 Select incorrect statement(s) for given carbonyl compounds.
 (A) $[\text{Tc}(\text{CO})_6]^+$ acts as reducing agent and $[\text{Nb}(\text{CO})_6]^-$ acts as oxidizing agent
 (B) $[\text{Nb}(\text{CO})_6]^-$ acts as reducing agent and $[\text{Tc}(\text{CO})_6]^+$ acts as oxidizing agent
 (C) “Nb–C” bond order in $[\text{Nb}(\text{CO})_6]^-$ is greater than “Tc–C” bond order in $[\text{Tc}(\text{CO})_6]^+$
 (D) “CO” bond order is greater in $[\text{Nb}(\text{CO})_6]^-$ than in $[\text{Tc}(\text{CO})_6]^+$

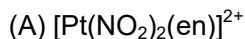
16. Which of the following statement(s) is/are correct?
- (A) The oxidation state of iron in sodium nitro prusside: $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ is +II
- (B) $[\text{Ag}(\text{NH}_3)_2]^+$ is linear in shape
- (C) In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridized
- (D) In $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ complex ion, Co is d^2sp^3 hybridized
17. Which of the following statements(s) is/are incorrect?
- (A) In $[\text{CoBrCl}(\text{en})_2]^+$ geometrical isomerism exists, while optical isomerism does not exist
- (B) Potassium aquadicyanosuperoxoperoxoperoxochromate(III) is IUPAC name of $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O}_2)(\text{H}_2\text{O})]$
- (C) There are 3 geometrical and 15 stereoisomers possible for $[\text{Pt}(\text{NO}_2)(\text{NH}_3)(\text{NH}_2\text{OH})(\text{py})]^+$ and $[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)(\text{py})]$ respectively
- (D) cis and trans-forms are not diastereomers to each other
18. In the crystal field of the complex $[\text{Fe}(\text{Cl})(\text{CN})_4(\text{O}_2)]^{4-}$, the electronic configuration of metal is found to be t_{2g}^6, e_g^0 then which of the following is/are true about this complex ion –
- (A) It is a paramagnetic complex
- (B) O – O bond length will be more than found in O_2 molecule
- (C) Its IUPAC name will be chlorotetracyanosuperoxidoferrate(II) ion.
- (D) It will show geometrical as well as optical isomerism

NUMERIC

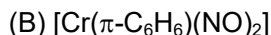
19. Total number of complexes among the following which are optically active?
- (i) $[\text{Cr}(\text{Ox})_3]^{3-}$ (ii) $[\text{Pt}(\text{Cl}_2)(\text{en})]$
- (iii) $\text{cis-}[\text{Rh}(\text{Cl}_2)(\text{NH}_3)_4]^+$ (iv) $[\text{Ru}(\text{dipy})_3]^{3+}$
- (v) $\text{cis-}[\text{Co}(\text{NO}_2)_3(\text{dien})]$ (vi) $\text{Trans-}[\text{Co}(\text{NO}_2)_3(\text{dien})]$
- (vii) $\text{cis-}[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$
20. Consider the following ligands NH_2^- , acac, OH^- , Gly, O_2^- , Phen, DMG, NO_2^- , CO_3^{2-} , Cl^- , CH_3COO^- , en, SO_4^{2-} . Then calculate value of “P + Q – R – S”
- Here
- P: Total number of ligands which act as bridging as well as monodentate only,
- Q: Total number of flexidentate ligands
- R: Total number of bidentate ligands only
- S: Total number of unsymmetrical bidentate ligands
21. How many of the following complex ions have dsp^2 hybridization $[\text{PdCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Pd}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$, Cis platin Zeise's salt, Wilkinson's catalyst, $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{Gly})_2]$, $[\text{Ni}(\text{CO})_4]$

MTC

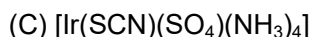
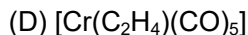
22. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with one or more than one entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

Column-I**Column-II****(Characteristics of complex ion compound)**

(P) Stable according to E.A.N. rule



(Q) Ligand acts as ambidentate

(R) Bond order of all M-L bond > 1.0 

(S) Bond order of all ligand decreases

(T) Co-ordination number of the central metal is six

(A) $A \rightarrow \text{P, R, T}; B \rightarrow \text{P, T}; C \rightarrow \text{Q, T}; D \rightarrow \text{Q, S, T}$ (B) $A \rightarrow \text{P, Q, T}; B \rightarrow \text{P, R, S}; C \rightarrow \text{P, Q, T}; D \rightarrow \text{P, R, S, T}$ (C) $A \rightarrow \text{R, T}; B \rightarrow \text{Q, T}; C \rightarrow \text{P, T}; D \rightarrow \text{S}$ (D) $A \rightarrow \text{P, Q, S}; B \rightarrow \text{Q, R, S}; C \rightarrow \text{P}; D \rightarrow \text{P, Q}$

23. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with one or more than one entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

Column-I**Column-II**

(A) Sodium nitroprusside

(P) $\mu = 0 \text{ BM}$

(B) Brown ring complex

(Q) Octahedral

(C) Complex of Ag formed during its extraction

(R) $\mu = 15 \text{ BM}$

(D) Potassium ferrocyanide

(S) NO^+ ligand(A) $A \rightarrow \text{P, R, T}; B \rightarrow \text{P, T}; C \rightarrow \text{Q, T}; D \rightarrow \text{Q, S, T}$ (B) $A \rightarrow \text{P, T}; B \rightarrow \text{P, R, S}; C \rightarrow \text{P, Q, T}; D \rightarrow \text{P, R, S, T}$ (C) $A \rightarrow \text{R, T}; B \rightarrow \text{Q, T}; C \rightarrow \text{P, T}; D \rightarrow \text{S}$ (D) $A \rightarrow \text{P, Q, S}; B \rightarrow \text{Q, S}; C \rightarrow \text{P}; D \rightarrow \text{P, Q}$

24. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with one or more than one entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

Column-I**Column-II**

- | | |
|--|---------------------------|
| (A) $\text{Na}_2[\text{Pt}(\text{SCN})_2(\text{ox})_2]$ | (P) Ionization isomerism |
| (B) $[\text{CrCl}_2(\text{NH}_3)_4]\text{NO}_3$ | (Q) Linkage isomerism |
| (C) $[\text{Pt}(\text{NO}_2)(\text{Gly})(\text{NH}_3)]$ | (R) Geometrical isomerism |
| (D) $\text{K}_3[\text{Fe}(\text{OH})_2(\text{C}_2\text{O}_4)_2]$ | (S) Optical isomerism |

(A) $A \rightarrow \text{P, R, T; } B \rightarrow \text{P, T; } C \rightarrow \text{Q, T; } D \rightarrow \text{Q, S, T}$

(B) $A \rightarrow \text{P, T; } B \rightarrow \text{P, R, S; } C \rightarrow \text{P, Q, T; } D \rightarrow \text{P, R, S, T}$

(C) $A \rightarrow \text{R, T; } B \rightarrow \text{Q, T; } C \rightarrow \text{P, T; } D \rightarrow \text{S}$

(D) $A \rightarrow \text{Q, R, S; } B \rightarrow \text{P, R; } C \rightarrow \text{Q, R; } D \rightarrow \text{R, S}$

SUBJECTIVE ANSWER TYPE

25. (A), (B) and (C) are three complexes of chromium (III) with the empirical formula $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. All the three complexes have water or chloride ion as ligands. Complex (A) does not react with concentrated H_2SO_4 , whereas complexes (B) and (C) lose 6.75% and 13.5% of their original weight, respectively, on treatment with concentrated H_2SO_4 .
- Identify (A), (B) and (C)
 - Write their formulae
 - Calculate their EAN.
 - By the addition of AgNO_3 what happens with each complex.

Answer Key CAPS-8

- | | | | | |
|-----------|------------|-----------|------------|------------|
| 1. (D) | 2. (C) | 3. (A) | 4. (D) | 5. (A) |
| 6. (B) | 7. (A) | 8. (C) | 9. (B) | 10. (B) |
| 11. (C) | 12. (B) | 13. (D) | 14. (A) | 15. (ABD) |
| 16. (ABD) | 17. (ABCD) | 18. (ABC) | 19. (2.00) | 20. (1.00) |
| 21. (7.0) | 22. (B) | 23. (D) | 24. (D) | |

